Lecture 07 — Fourier Series

MATH 2201 · Week 4

Any periodic function f(x) with period 2L can be represented as an infinite sum of sines and cosines. This representation is called the Fourier series of f.

    f(x) = a0/2 + SUM[an cos(n pi x / L) + bn sin(n pi x / L)]

The coefficients are obtained by exploiting orthogonality of the basis functions over one full period. Multiplying both sides by cos(m pi x / L) and integrating term by term collapses the sum to a single surviving term.

    an = (1/L) INTEGRAL[-L..L] f(x) cos(n pi x / L) dx
    bn = (1/L) INTEGRAL[-L..L] f(x) sin(n pi x / L) dx

Convergence is guaranteed at every point of continuity by the Dirichlet theorem. At a jump discontinuity the series converges to the midpoint of the jump, and the partial sums overshoot near the jump by roughly 9% of the gap. This is the Gibbs phenomenon and it does not vanish as more terms are added.